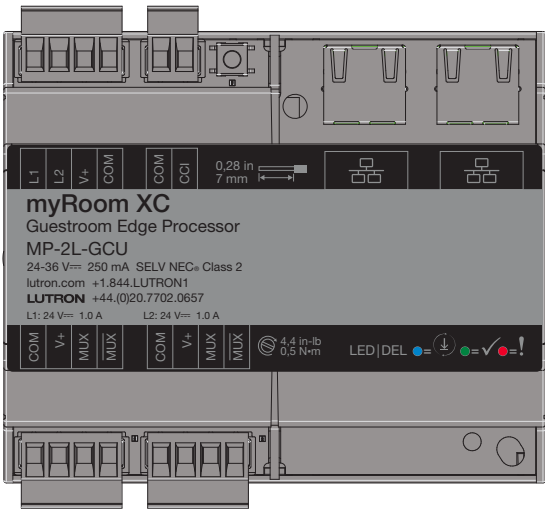


Guestroom Edge Processor

The guestroom processor coordinates lighting, shades/draperies, and temperature control in hotel guestrooms. These technologies can be automated for guest comfort when the room becomes occupied and for energy savings when the room is vacant. The guestroom processor can integrate with third-party systems via the onboard Ethernet link. Examples include Property Management Systems (PMS), Central Electronic Locking Systems (CELS), Service Management Systems (SMS), and Building Management Systems (BMS). The Ethernet link also allows for connection to the Lutron Designer software and Lutron dashboard, and communication between multiple guestroom processors. All processors on a project must be connected to a single network.

Features

- Designed to control, manage, and monitor Lutron DIN power modules, QS and Pico wallstations, Sivoia and Contract Roller QS shades, and QS drapery systems.
 - The myRoom XC system brings switching, dimming, motorized window shades, digital LED drivers, and smart sensors together under one software tool.
- Ships with mounting bracket for ease of installation into existing Lutron panels.
 - Supports both astronomic and time-of-day events to automatically control the lights and shades/draperies in the guestroom.



MP-2L-GCU shown
(COM/CCI terminal is not functional)

Guestroom Processor Model	Limitations per Processor			Limitations per QS Link		
	Number of QS Links	Number of Ethernet Ports	QS Device Count	Keypad Count ¹	Occupancy Sensor Count	Switchleg Count
MP-1L-GCU	1	2	50 (per link)	50	50	256
MP-2L-GCU	2	2	99 (per link)	100	100	512

¹ Keypads also include Pico wireless controls connected via a QSM.

Specifications

Regulatory Approvals

- cULus
- NOM
- CE Compliant
- UKCA Compliant
- FCC
- IC

Power

- Processor: 24–36 V $\overline{=}$ 250 mA
- Links: 24–36 V $\overline{=}$ 1 A per link

Environment

- Operating temperatures 32 °F to 104 °F (0 °C to 40 °C)
- Relative humidity: 0% to 90% non-condensing
- For indoor use only

Typical Power Consumption

- 4.2 W; 8 Power Draw Units (PDUs)
Test conditions: Two Ethernet links connected, both device links in use

ESD Protection

- Tested to withstand electrostatic discharge without damage or memory loss, in accordance with IEC 61000-4-2

Surge Protection

- Tested to withstand surge voltages without damage or loss of operation, in accordance with IEEE C62.41-1991 Recommended Practice for Surge Voltages in Low-Voltage AC Power Circuits

System Capacity

- A maximum of 16 guestroom processors per system

Power Failure

- **10-year power failure memory:** Should the power be interrupted, the processor will return to its previous state prior to the interruption when power is restored

Mounting

- Mounts into the enclosures listed on pages 9 and 11 using supplied mounting bracket and two supplied screws or in third-party DIN enclosures

Low-Voltage Link Wire Type

- Two pairs — one pair 18 AWG (1.0 mm²), one pair 22 AWG to 18 AWG (0.5 mm² to 1.0 mm²) twisted and shielded — IEC PELV/NEC® Class 2 cable

Internet Connection

- Providing the guestroom processor with an internet connection is highly recommended for all projects and applications. The following myRoom XC features require an internet connection:
 - Automatic firmware updates
 - DALI® emergency testing functionality
 - Remote diagnostics, service, and support
 - Lutron dashboard
- This internet connection is outbound from the guestroom processor to the cloud. See the Commercial Lighting Control System IT Implementation Guide (Lutron P/N 040453) at www.lutron.com for details.

Job Name:	Model Numbers:	
Job Number:		

Specifications *(continued)*

System Capabilities

- Lutron QS Link allows control and programming of:
 - DIN power modules
 - Palladiom and seeTouch QS keypads
 - Palladiom QS thermostat
 - Pico wireless controls (through QS sensor module)
 - Sivoia QS and Contract Roller QS shades, and QS drapery systems
 - Interfaces for door sensors, window sensors, room chimes, and privacy and service controls
- Keypad buttons can be programmed to toggle lights, select room scenes, or raise and lower lighting or shades / draperies.
- Guestroom processors interface with the hotel Property Management System (PMS), Central Electronic Lock System (CELS) and Hotel Service Management System (SMS) via the Lutron Hotel Integration Appliance. Please refer to specification submittal (P/N 3691261) at www.lutron.com for more information.
- Guestroom processors interface with the hotel Building Management System (BMS) via a wired Ethernet hotel network connection. Software license LMRXC-BAC-PR required for native BACnet® interface.
- BACnet® interface allows BMS to monitor and control room lighting, temperature, and room occupancy status.
- Lighting and HVAC load schedule data can be programmed at start-up. This data is used to calculate room lighting and HVAC power usage, and is available to the hotel BMS for logging and reporting via the native BACnet® interface.
- Lutron dashboard provides a powerful software to monitor, analyze and report on the myRoom XC system.

Manual DALI® Emergency Testing

- System capability that enables the guestroom processor to run and report test results of DALI®-2 certified emergency loads connected to the QSN-2DALUNV-D/S, QSN-1DALUNV-D modules.
- The guestroom processor must be connected to the internet to obtain the test results report
 - User must initiate test through the dashboard

User Access

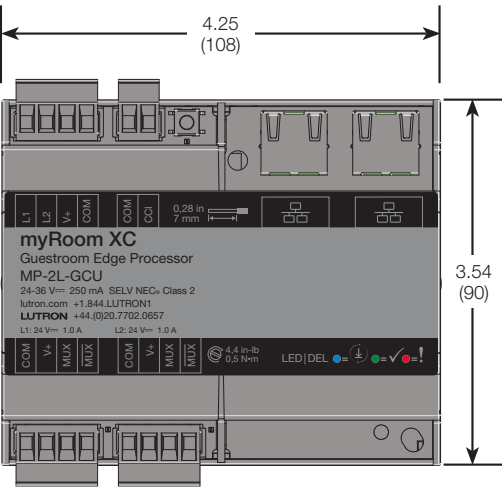
- myLutron username and password required for user access

Job Name: 	Model Numbers: 	
Job Number:		

Dimensions: MP-1L-GCU, MP-2L-GCU

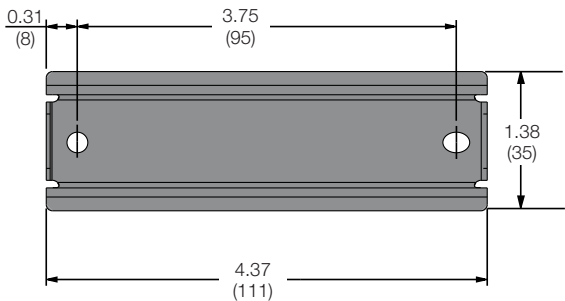
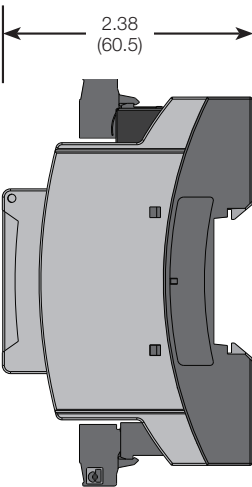
All dimensions are shown as: in (mm)

Front View



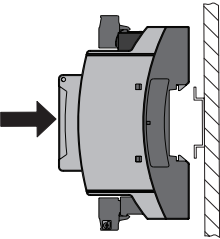
MP-2L shown

Side View

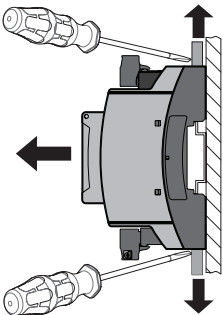


Note: Includes (2) 8-32 x .25 in (6.35 mm) pan head machine screws for mounting.

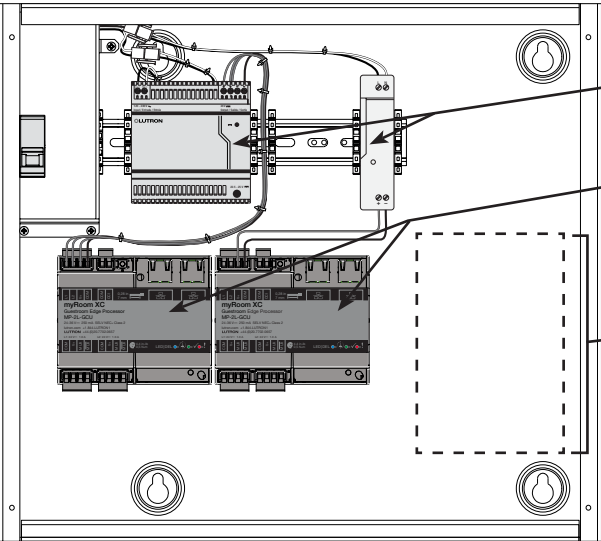
Processor Mounting



Installing processor onto mounting bracket/DIN rail

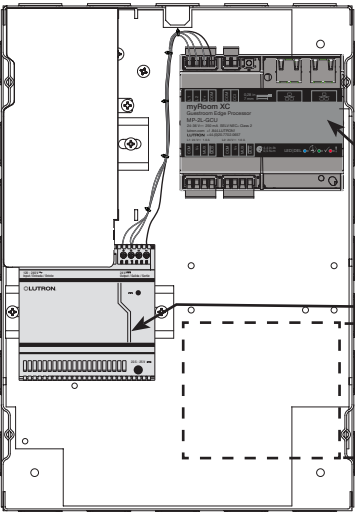


Removing processor from mounting bracket/DIN rail



LV16

- Power Supply¹
QSPS-DH-1-75 or MQSPS-DH-1-30
(up to 2)
- Guestroom Processor^{1,2}
MP-1L-GCU or MP-2L-GCU
OR
System Interfaces^{1,2}
(see pages 9 and 11 for complete list)
- Location for mounting QS Wire Landing Board¹
QS-WLB
(up to 4)



L-LV14

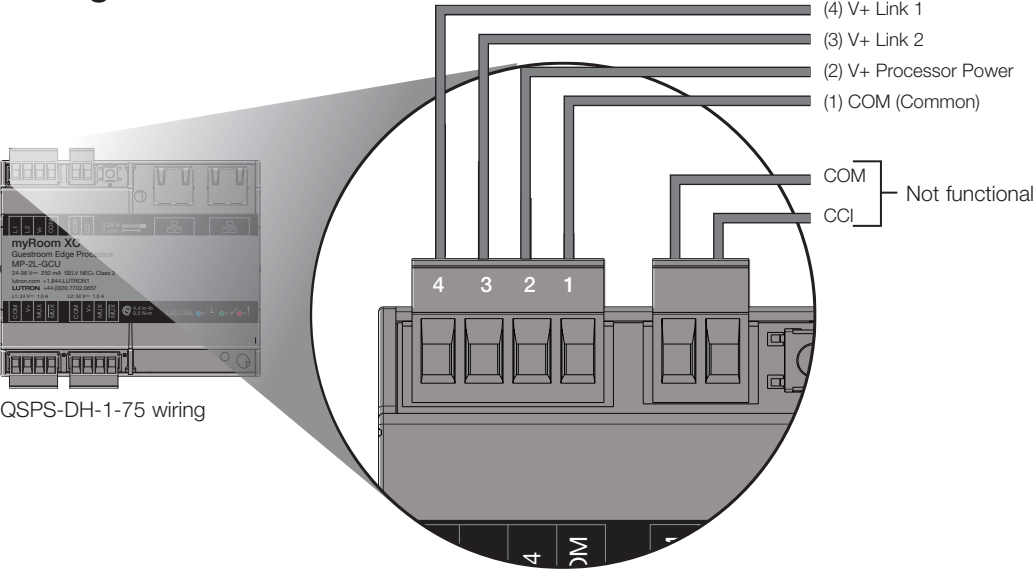
- Guestroom Processor¹
MP-1L-GCU or MP-2L-GCU
- Power Supply¹
QSPS-DH-1-75 or MQSPS-DH-1-30³
- Location for mounting QS Wire Landing Board¹
QS-WLB
(up to 2)

¹ All components sold separately.
² Maximum of two control units/interfaces allowed per enclosure.
³ Requires mounting kit MQSPS-BRK.

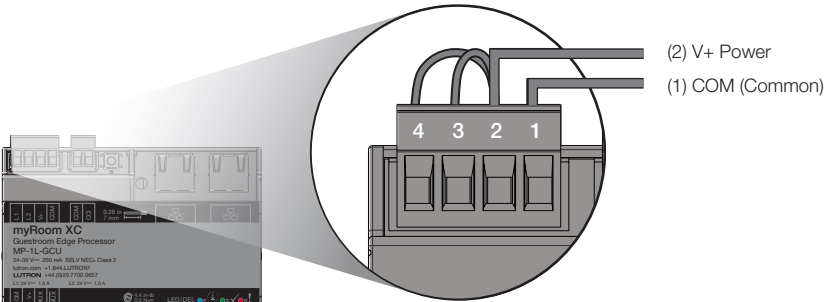
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Job Name:	Model Numbers:	
Job Number:		

Wiring: Power

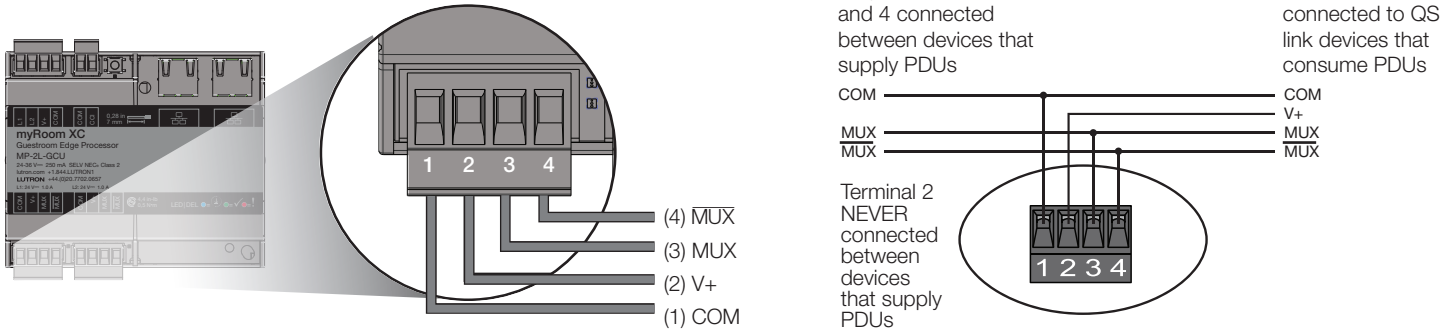


QSPS-DH-1-75 wiring



MQSPS-DH-1-30 wiring

Wiring: QS Link (all models)



QS Link Wiring Options

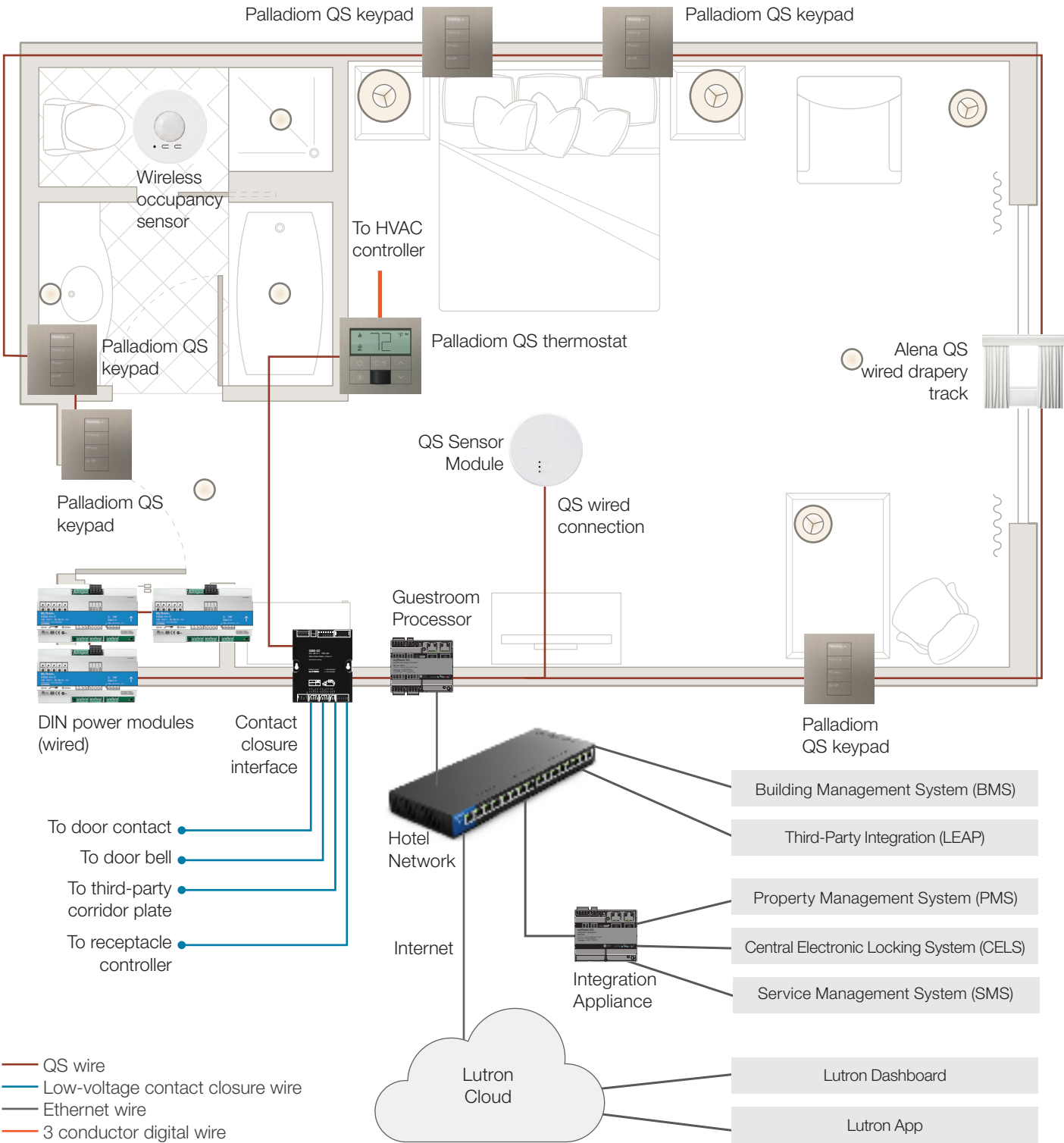
QS Link Wiring Length	Wire Gauge	Available from Lutron in one cable: *
Less than 500 ft (153 m)	Power (terminals 1 and 2): 1 pair 18 AWG (1.0 mm²)	QS-CBL-LSZH (low-smoke zero-halogen)
	Data (terminals 3 and 4): 1 pair 22 AWG (0.5 mm²), twisted and shielded	GRX-CBL-346S (non-plenum) GRX-PCBL-346S (plenum)
500 ft to 2000 ft (153 m to 610 m)	Power (terminals 1 and 2): 1 pair 12 AWG (4.0 mm²)	GRX-CBL-46L (non-plenum)
	Data (terminals 3 and 4): 1 pair 22 AWG (0.5 mm²), twisted and shielded	GRX-PCBL-46L (plenum)

* Varies by region, refer to the cable spec.

Notes

- Link communication uses IEC PELV/NEC® Class 2 low-voltage wiring.
 - Follow all local and national electrical codes when installing IEC PELV/NEC® Class 2 wiring with line voltage/mains wiring.
 - Terminals accept:
 - One 22 AWG to 12 AWG (0.5 mm² to 4.0 mm²) wire
 - Up to two 18 AWG (1.0 mm²) wires
 - The total wire length of a link must not exceed 2000 ft (610 m)
- The guestroom processor provides 33 Power Draw Units (PDUs) on each QS Link when using a QSPS-DH-1-75 power supply. For more information, see “Power Draw Units on the QS Link” (Lutron P/N 369405) at www.lutron.com
 - QS Link Wiring can be T-tapped or daisy-chained.
 - Terminal 2 (V+) should NEVER be connected between devices that supply PDUs.
 - For QS Link power supply wiring connection details, refer to the installation instructions for the specific power supply model being used.

System Diagram



Security Statement

Lutron takes cybersecurity very seriously. We actively monitor the threat landscape and take a proactive approach to security and privacy, continuously working to update and enhance our systems and processes.

At Lutron, we call our approach to cybersecurity “**Secure Lifecycle**”, and we would like to present the following steps we take to protect your security and privacy:

- **Security by Design.** When building a new system, Lutron utilizes a dedicated security team to ensure best practices are implemented. Security is built in. It is not an afterthought or an add-on.
- **Third-Party Validation.** Security is complicated. Lutron has a dedicated team of internal experts, but we also leverage external experts to double-check our work, and to make security recommendations.
- **Continuous Monitoring and Improvements.** Security is a constantly moving target. Lutron uses a dedicated security team to continuously monitor for potential threats and, when needed, send out security patches to update installed systems.
- **Ongoing Support.** Lutron has the resources you need to answer questions about security when they arise.

We incorporate a variety of security features into our product designs. These features include recommendations from the National Institute of Standards and Technology (NIST) among others, and they are aimed at meeting our secure lifecycle protections. While we do not publish a comprehensive list of our security features, the following list is a small example of some of the techniques employed in our system designs for Processors, and associated services (such as mobile applications and cloud resources):

1. Secure and authenticated remote access with unique keys for every system.
2. A secure hardware element (“chip”) on all processors to guard the keys used for secure communication and authentication.
3. Enforcing industry-standard encrypted communication and techniques for our integration protocols to the highest extent possible. Any integrated third-party components or systems should be evaluated independently.
4. Secure commissioning – all communication between the system programming software tool/app and the processors is encrypted and authenticated. Programming a system requires permission to access that system.
5. Security updates are pushed out automatically to the lighting system for urgent security patches. Lutron is committed to one year of security support from system start-up date.
6. Use of industry-standard techniques for cloud-based integrations, such as OAuth2.0
7. Signed processor firmware to ensure a firmware update is authentically from Lutron.

If you have additional questions or would like to make a vulnerability disclosure to Lutron, please contact Lutron’s 24/7 Technical Support Line at 1.844.LUTRON1 or email us at support@lutron.com.

Job Name:	Model Numbers:	
Job Number:		

Compatible Models - North America

Load Controls

- QSN-4T5-120-D
- QSN-4T20-D
- QSN-4S20-D
- QSN-1DALUNV-D
- QSN-2DALUNV-D
- QSN-4S8-120-D
- QSN-4A5-120-D
- MQSE-2S1-D
- MQSE-3S1-D
- MQSE-4S1-D
- MQSE-2A1-D
- MQSE-3A1-D
- MQSE-4A1-D

Wall Controls

- Px2-*
- PX-*
- QSWA-*
- QSWAS-*
- QSWE-*
- QSWS2-*
- QSWP-CP-*
- QSWP-DM-*
- MWP-*

Shades

- Contract Roller Shades
- Sivoia QS Roller Shades and Drapery
- Alena QS Drapery

Sensors

- LOS-*
- LRF2-OCR2B*
- LRF2-OKLB*
- LRF2-OWLB*
- LRF2-OHLB*
- LUT-WS*
- QSM2-*

Integration Interfaces

- QSE-IO
- QSE-CI-WCI

Power Supplies

- QSPS-*
- MQSPS-DH-1-30

Thermostat/Fan Coil Unit Controller

- MWP-T-OHW*
- SMC53-MYRM
- SMC55-MYRM

Enclosures

- LV16-120
- L-LV14-120
- HQ-LV21-120
- PD4-36F-120
- PD8-59F-120
- LUT-8x8-ENC

* Designates additional model number characters that may vary depending on the specific model chosen.

Job Name:	Model Numbers:	
Job Number:		

Coordinating Accessories - North America

Accessories

- LUT-19AV-1U
- L-PED*
- PICO-*

Receptacles and Wallplates

- NTR-15-HDTR-XX
- NTR-15-DDTR-XX
- SCR-15-HDTR-XX
- SCR-15-DDTR-XX
- CAR-15-HDTR-XX
- CAR-15-DDTR-XX
- RP-FDU-10-*
- NTR-15-TR-*
- NTR-20-TR-*
- NTR-15-GFST-*
- NTR-20-GFST-*
- NTR-15-UBTR-*
- SCRS-15-TR-*
- SCRS-20-TR-*
- SCR-15-GFST-*
- SCR-20-GFST-*
- SCR-15-UBTR-*
- CARS-15-TR-*
- CAR-15-GFST-*
- CAR-15-UBTR-*
- NT-R3-NFB-*
- NT-R3R3-FB-*
- NT-R3R3R3-FB-*
- NT-R3R3R3R3-FB-*
- LFGx-*
- LTR-*
- LWT-*
- LPFP-*
- CW-*
- SC-*

* Designates additional model number characters that may vary depending on the specific model chosen.

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Page 11

Job Name:	Model Numbers:		
Job Number:			

Compatible Models - 220-240 V~ Regions

Load Controls

- QSNE-4A5-230-D
- QSNE-4S10-D
- QSNE-4T10-230-D
- QSN-1DALUNV-D
- QSN-2DALUNV-D
- QSNE-4S5-230-D
- MQSE-2S1-D
- MQSE-3S1-D
- MQSE-4S1-D
- MQSE-2A1-D
- MQSE-3A1-D
- MQSE-4A1-D

Wall Controls

- Px2-*
- PX-*
- QSWA-*
- QSWAS-*
- QSWE-*
- QSWS2-*
- QSWP-CP-*
- QSWP-DM-*
- MWP-*

Shades

- Contract Roller Shades
- Sivoia QS Roller Shades and Drapery
- Alena QS Drapery

Sensors

- LOS-*
- LRFx-OCR2B*
- LRFx-OKLB*
- LRFx-OWLB*
- LRFx-OHLB*
- LUT-WS*
- QSMx-*

Integration Interfaces

- QSE-IO
- QSE-CI-WCI

Power Supplies

- QSPS-*
- MQSPS-DH-1-30

Thermostat/Fan Coil Unit Controller

- MWP-T-OHW*
- MWP-T-OHW-QB-A
- SMC53-MYRM
- SMC55-MYRM

Enclosures

- L-LV14-230
- L-LV21-230
- PD4-36-BU10-2L4
- PD8-64-BU10-2L4
- PD8-64-F
- LUT-8x8-ENC

* Designates additional model number characters that may vary depending on the specific model chosen.

Job Name:	Model Numbers:	
Job Number:		

Coordinating Accessories - 220-240 V~ Regions

Accessories and Wallplates

- LUT-19AV-1U
- LUT-5x10-ENC
- LFGx-*
- LWT-*
- LPFP-*
- L-PED*
- PICO-*
- CW-*
- SC-*

* Designates additional model number characters that may vary depending on the specific model chosen.

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Job Name:	Model Numbers:	
Job Number:		